

Continuum: A System for Agentic Multi-turn Image Generation001

Supplementary Material002

This PDF accompanies the anonymous supplementary repository. See README.md for the code, result browser, and machine-readable scores distributed alongside it.003004

Appendices005

A. Additional Quantitative Results Published comparisons, category and compositionality breakdowns, and trace statistics.006007

B. Qualitative Examples A selected subset of qualitative examples aiming for prompt coverage. Only the final output is included for the agentic loop.008009

C. Tool Schemata Compact parameter summaries; complete MCP definitions are included in code/tool-schemas.json.010

D. Instructions and System Prompts The instructions included with the MCP server (which are appended to the system prompt in most harnesses) as well as AGENTS.md that the batch harness used to instruct the agent in our UniGen-Bench++ [7] experiments.011012013

Agent	Generator	Score	Place
	GPT Image 1.5	95.77	1
Kimi K3	Qwen-Image	94.60	-
Qwen3.8-27B	FLUX.2-dev	94.40	-
Kimi K3	FLUX.2-dev	94.14	-
Qwen3.8-27B	Qwen-Image	93.60	-
Kimi K3	FLUX.2-klein-9B	93.43	-
	Nano Banana Pro	92.72	2
Qwen3.8-27B	FLUX.2-klein-9B	92.49	-
	GPT Image	92.48	3
	Imagen-4.0-Ultra-preview-06-06	91.65	4
	FLUX-2-max	90.85	5
	Seedream-4-5-251128	89.70	6
	FLUX-2-pro	88.35	8
	Nano Banana	87.29	10
	<i>FLUX.2-dev</i>	84.76	12
	<i>Qwen-Image</i>	78.36	16
	<i>FLUX.2-klein-9b</i>	78.28	17

Table 1. **UGB++ published context.** Overall is the unweighted mean of ten primary-dimension accuracies, on a 0–100 scale. Published scores and places are the leaderboard snapshot used for this comparison; omitted rows account for gaps in place numbers. Bold rows are our agentic runs, not claims of an official leaderboard position. Italics highlight published direct baselines for the generators we use. Our measured direct baselines used in the category breakdowns are 83.35 for FLUX.2-dev, 79.41 for Qwen-Image, and 78.94 for Klein 9B.

014 **A. Additional Quantitative Results**

015 **A.1. Evaluation Settings**

016 The main quantitative comparison uses Kimi K3 and Qwen3.8-27B with FLUX.2-klein-9B, FLUX.2-dev, and Qwen-Image.
017 We use our own direct baselines, with four samples per prompt and a fixed 1024×1024 output size. Agent runs can generate,
018 inspect, edit, segment, and inpaint, with a limit of 100 agent turns per episode. FLUX.1-Fill is used for inpainting. A
019 run can be retried after an endpoint error; the recorded trace counts below count only the latest saved transcript for each
020 prompt/sample.

021 The Qwen-Image arm uses Qwen-Image for generation and Qwen-Image-Edit-2511 for editing. The completed UGB++
022 direct FLUX.2-dev run used the dual-GPU BF16 path, while its agent workers used single-GPU FP8 weight-only quanti-
023 zation. The GenEval 2 FLUX.2-dev direct and agent workers both use the same single-GPU FP8 weight-only path. These
024 settings are relevant when interpreting differences between generators or comparing against published implementations.

025 **A.2. Published Comparisons**

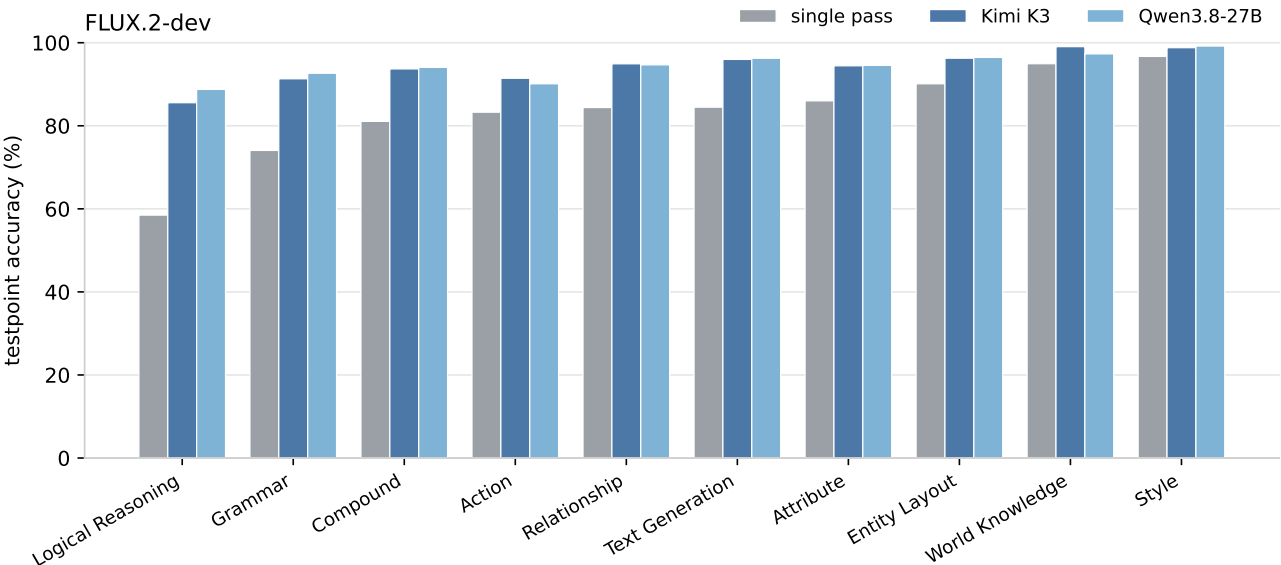
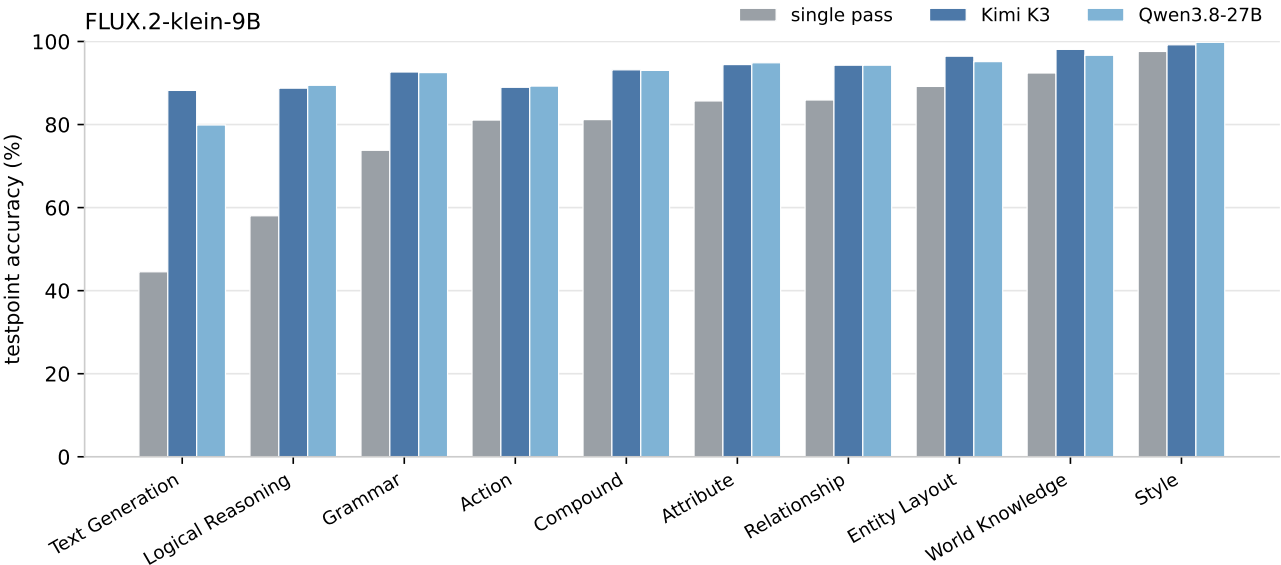
026 The following tables provide context from published evaluations. UGB++ uses the mean of ten primary-dimension accura-
027 cies. GenEval 2 uses the Soft-TIFA GM and AM metrics [3]. These comparisons use different agent, precision, and compute
028 settings; published ranks are not assigned to our runs.

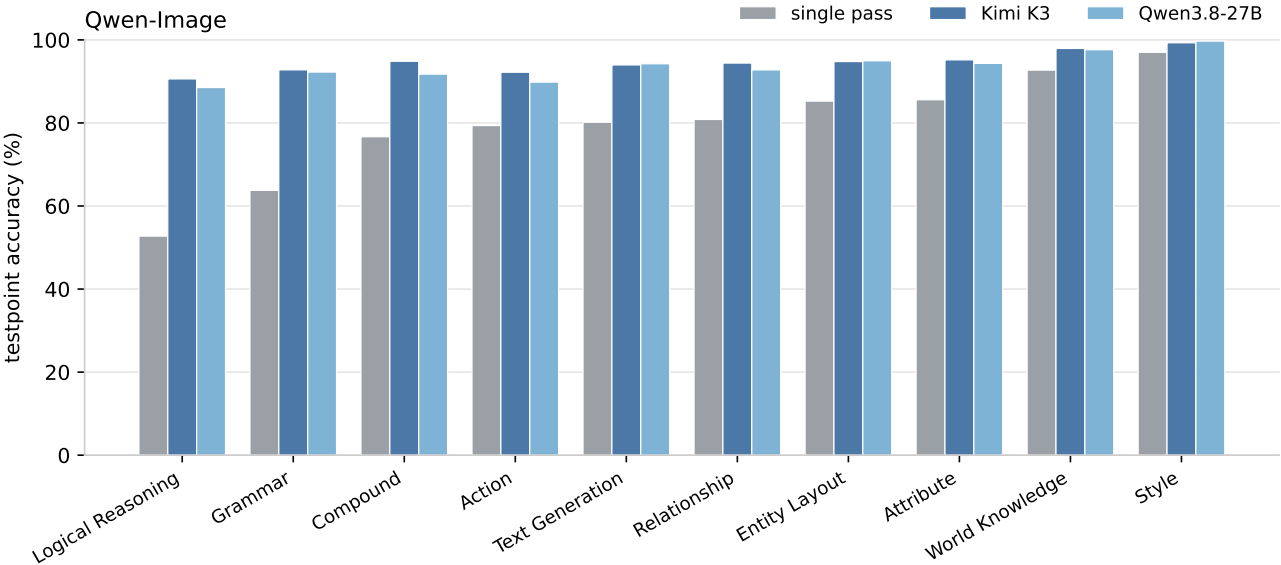
Method / Agent	Base generator	GM $\uparrow$	AM $\uparrow$
Kimi K3	Qwen-Image	79.76	93.25
Kimi K3	FLUX.2-dev	78.68	92.78
Qwen3.8-27B	Qwen-Image	76.53	92.44
Kimi K3	FLUX.2-klein-9B	75.39	91.62
Qwen3.8-27B	FLUX.2-dev	73.43	91.32
EPIC [†]	FLUX.2-klein-9B	71.46	90.25
Qwen3.8-27B	FLUX.2-klein-9B	71.28	90.45
RAISE [†]	FLUX.2-klein-9B	52.23	86.28
Direct [‡]	Gemini 2.5 Flash Image	44.60	82.80
Direct [†]	<i>FLUX.2-dev</i>	42.09	83.34
BoN+NVILA [†]	FLUX.2-klein-9B	41.60	82.38
Direct (ours)	<i>Qwen-Image</i>	40.25	82.89
Direct (ours)	<i>FLUX.2-dev</i>	39.80	81.60
T2I-Copilot [†]	FLUX.2-klein-9B	36.56	81.19
Direct (ours)	<i>FLUX.2-klein-9B</i>	34.57	79.25
Direct [†]	<i>FLUX.2-klein-9B</i>	34.16	79.06
Direct [‡]	Qwen-Image	33.80	80.80
CoT [‡]	BAGEL	23.10	70.90
Direct [‡]	SD3.5-large	22.80	69.70
Direct [‡]	SD3-medium	21.30	66.30
Direct [‡]	FLUX.1-dev	21.10	67.10
Direct [‡]	SDXL	9.10	50.10
Direct [‡]	SD2.1	5.20	40.80

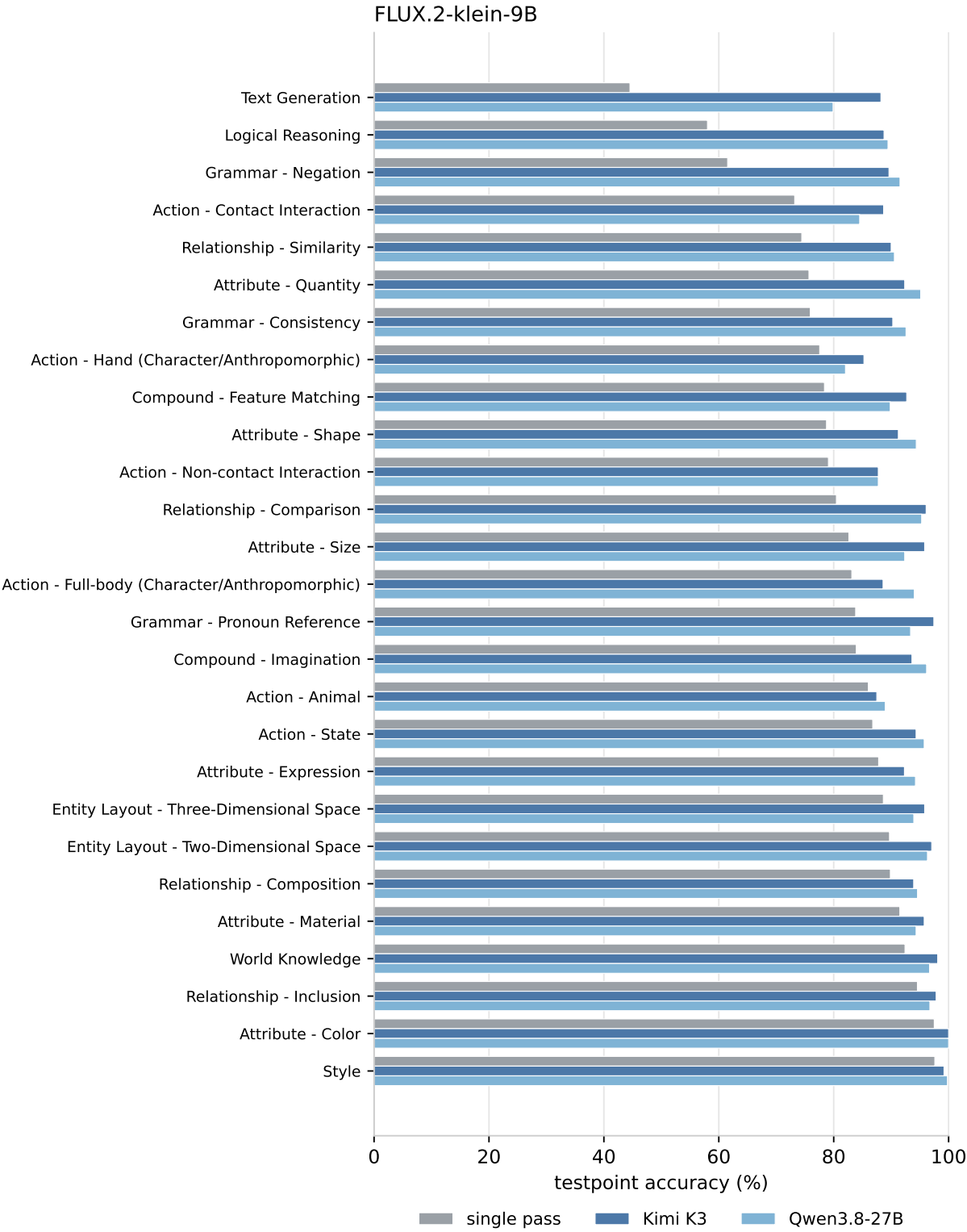
Table 2. **GenEval 2 Results.** Soft-TIFA geometric (GM) and arithmetic (AM) mean scores on a 0–100 scale, ordered by GM. **Bold** rows use our agentic loop; *italics* identify direct baselines. All our configurations average four samples per prompt over all 800 prompts. [†]: published results from EPIC [5], Tables 1 and 6 ($T = 0$). [‡]: published Soft-TIFA results from GenEval 2 [3], Table 4 (original prompts). Published comparisons use different agent and compute settings; our agent arms also allow FLUX.1-Fill inpainting. These are cross-paper comparisons, with no official leaderboard ranks assigned.

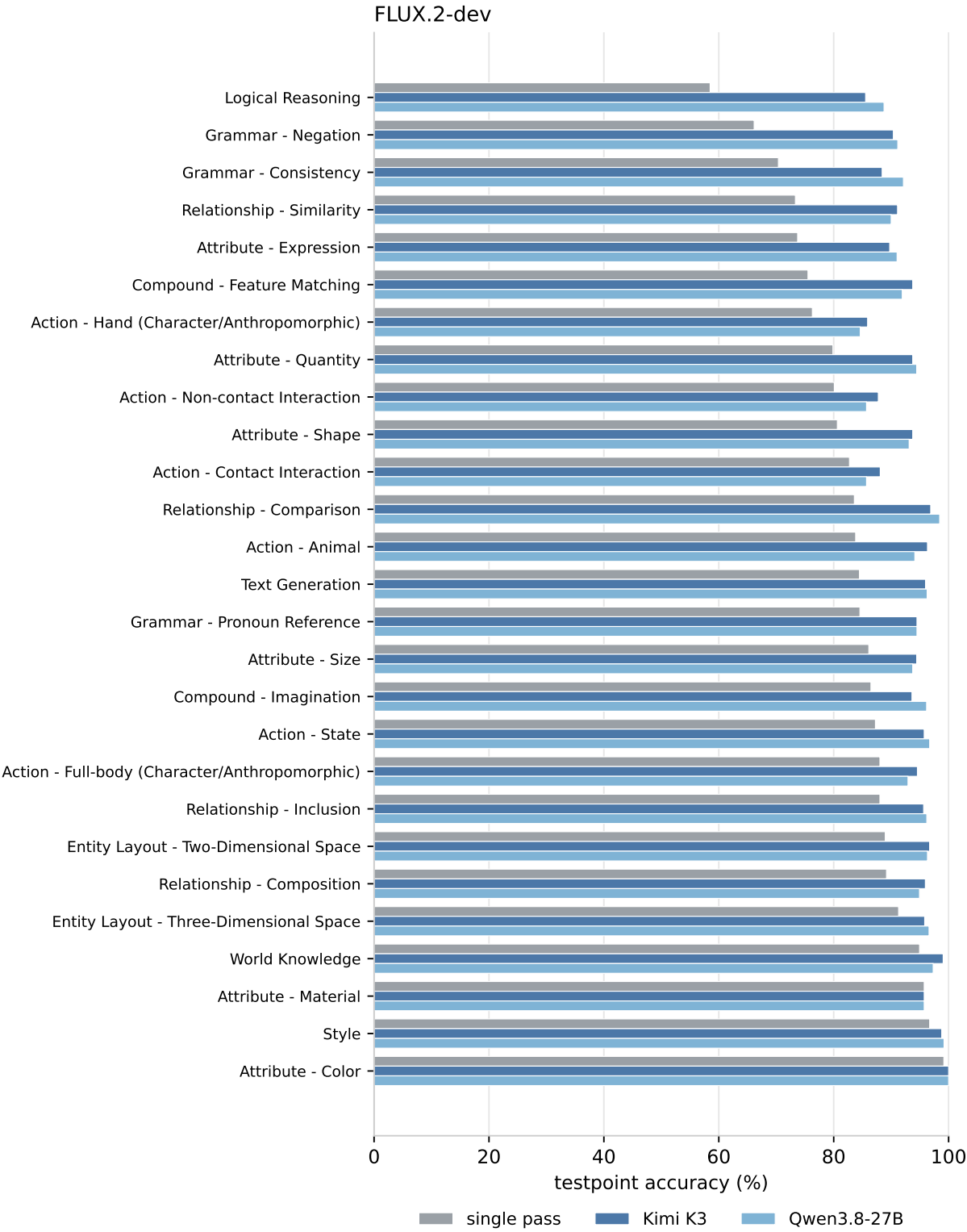
029 A.3. UGB++ Category Breakdowns

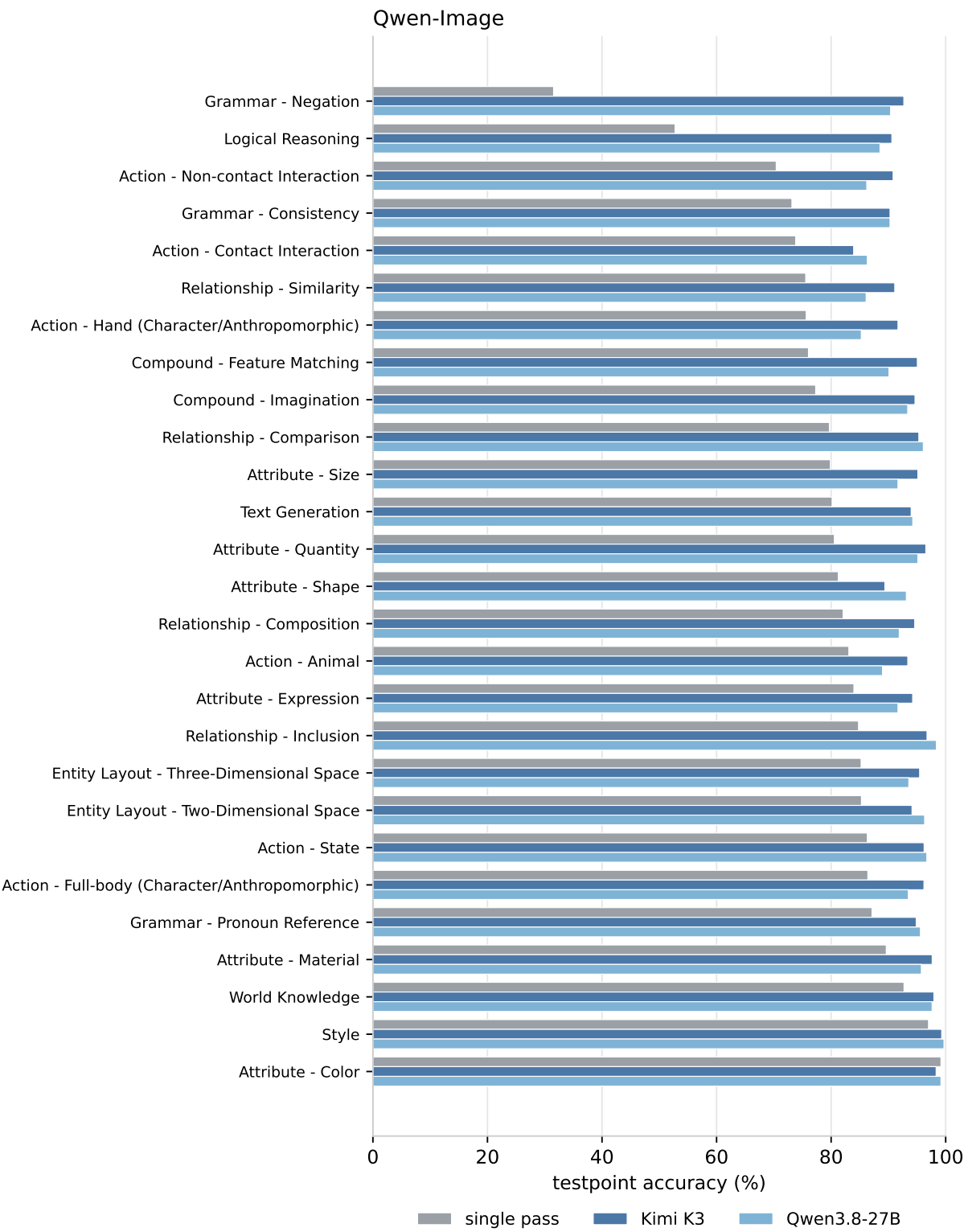
030 UniGenBench++ [7] testpoint accuracy by primary dimension and sub-dimension. Each panel compares our direct baseline
031 with Kimi K3 and Qwen3.8-27B for the same generator.











Generator / Agent	Object	Attribute	Count	Position	Verb
FLUX.2-klein-9B / Direct	95.58	88.79	58.11	74.78	37.05
FLUX.2-klein-9B / Kimi K3	96.31	96.06	85.74	87.00	49.36
FLUX.2-klein-9B / Qwen3.8-27B	96.02	95.72	83.32	85.39	45.58
FLUX.2-dev / Direct	96.20	92.11	63.63	70.40	43.35
FLUX.2-dev / Kimi K3	97.01	97.10	87.95	87.21	51.53
FLUX.2-dev / Qwen3.8-27B	96.65	96.30	85.27	85.07	49.42
Qwen-Image / Direct	98.74	88.08	69.68	60.46	39.78
Qwen-Image / Kimi K3	97.84	97.22	89.18	84.96	51.30
Qwen-Image / Qwen3.8-27B	97.75	96.45	87.45	84.44	51.58

Table 3. **GenEval 2 by skill.** Mean per-question soft score, averaged over four samples per prompt. Only fully scored configurations are included.

Generator / Agent	3	4	5	6	7	8	9	10
FLUX.2-klein-9B / Direct	63.79	51.00	41.43	36.25	26.67	22.82	20.16	14.41
FLUX.2-klein-9B / Kimi K3	89.80	86.57	81.75	77.38	72.02	70.53	65.36	59.69
FLUX.2-klein-9B / Qwen3.8-27B	86.35	83.68	80.44	72.81	67.43	63.73	61.21	54.55
FLUX.2-dev / Direct	67.75	61.85	47.09	45.92	26.70	25.91	22.36	20.85
FLUX.2-dev / Kimi K3	90.23	88.34	84.16	81.45	77.65	74.57	68.57	64.48
FLUX.2-dev / Qwen3.8-27B	86.03	84.38	80.76	77.07	70.60	67.52	63.10	57.97
Qwen-Image / Direct	74.35	62.37	48.53	46.93	28.56	27.99	17.98	15.30
Qwen-Image / Kimi K3	90.05	90.09	87.49	81.36	78.95	75.16	70.39	64.58
Qwen-Image / Qwen3.8-27B	89.19	88.22	83.20	80.15	75.00	70.85	64.41	61.21

Table 4. **GenEval 2 by compositionality.** Soft-TIFA GM for prompts containing 3–10 visual atoms, averaged over four samples. Only fully scored configurations are included.

A.4. GenEval 2 Skill and Compositionality Breakdowns

All nine configurations are summarized below. GM is sensitive to a low score on any required question, whereas per-skill AM measures individual visual requirements.

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Benchmark	Generator	Agent	Traces	Extra attempts	Mean turns	Image calls	Reasoning (%)
UGB++	FLUX.2-klein-9B	Kimi K3	2,400	1	5.26	1.98	85.7
GenEval 2	FLUX.2-klein-9B	Kimi K3	3,200	524	9.71	4.63	97.6
UGB++	FLUX.2-klein-9B	Qwen3.8-27B	2,400	8	6.06	1.92	100.0
GenEval 2	FLUX.2-klein-9B	Qwen3.8-27B	3,200	22	9.30	3.91	100.0
UGB++	FLUX.2-dev	Kimi K3	2,400	5	4.40	1.54	86.5
GenEval 2	FLUX.2-dev	Kimi K3	3,200	22	10.42	4.97	97.8
UGB++	FLUX.2-dev	Qwen3.8-27B	2,400	13	5.50	1.64	100.0
GenEval 2	FLUX.2-dev	Qwen3.8-27B	3,200	36	10.19	4.11	100.0
UGB++	Qwen-Image	Kimi K3	2,400	0	5.56	2.06	97.7
GenEval 2	Qwen-Image	Kimi K3	3,200	5	7.83	3.56	97.0
UGB++	Qwen-Image	Qwen3.8-27B	2,400	5	5.82	1.86	100.0
GenEval 2	Qwen-Image	Qwen3.8-27B	3,200	32	7.63	3.00	100.0

Table 5. **Recorded traces and interaction cost.** Each prompt/sample contributes its latest saved transcript once. Extra attempts count additional manifest records, including retries, whose transcripts may have been overwritten. Mean turns and image calls describe the retained transcripts: recorded assistant messages and successful generation, editing, and inpainting responses, respectively. Reasoning is the percentage of assistant messages with a nonempty returned reasoning field. These counts exclude request-level API retries and do not establish downstream training effectiveness.

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A.5. Recorded Traces and Interaction Cost

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We audited the latest saved transcript for each prompt/sample across all configurations: 33,600 agent trajectories containing 254,537 assistant turns and 103,788 successful image-model calls. The counts below describe the available interaction records, including returned reasoning where exposed by the endpoint. They are not counts of independently validated training examples, and no downstream model is trained on these traces in this work.

B. Qualitative Examples

Here we show a subset of qualitative examples. We encourage reviewers to view `site/index.html` for detailed traces. Each example shows a single prompt. The first row shows FLUX.2 Klein [2] and the second row shows FLUX.2 dev [1]. The “single pass” (left) column shows the result of passing the prompt to the generator directly.



single pass
✗ 0/1 Action - Full-body
(Character/Anthropomorphic)



Kimi K3
✓ 1/1
FLUX.2-klein-9B



Qwen3.8-27B
✓ 1/1



single pass
✓ 1/1

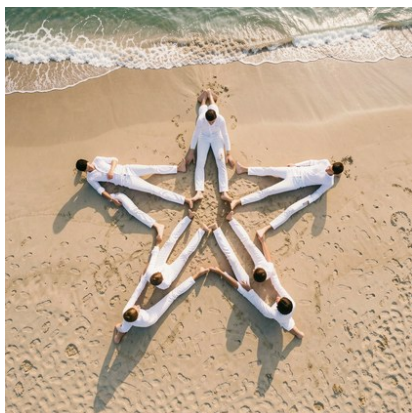


Kimi K3
✓ 1/1
FLUX.2-dev



Qwen3.8-27B
✗ 0/1 Action - Full-body
(Character/Anthropomorphic)

Figure 1. Please draw: A person made of building blocks is tiptoeing, trying to put a fallen star back on the missing corner of the moon.



single pass
✗ 0/2 Attribute - Shape, Entity Layout -
Two-Dimensional Space



Kimi K3
✓ 2/2

FLUX.2-klein-9B



Qwen3.8-27B
✓ 2/2

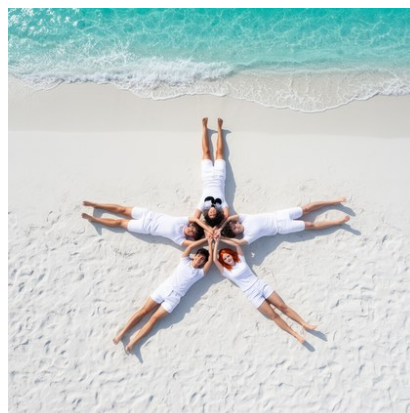
051



single pass
✗ 1/2 Attribute - Shape

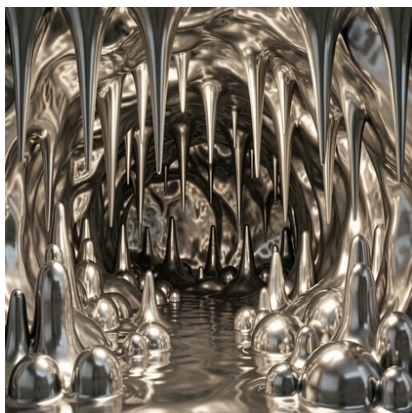


Kimi K3
✓ 2/2
FLUX.2-dev

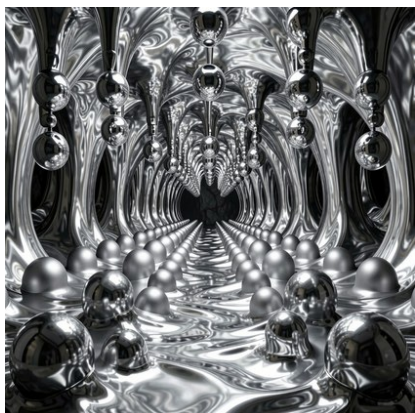


Qwen3.8-27B
✓ 2/2

Figure 2. Aerial photography shows five young people in white clothes forming a huge five-pointed star shape with their bodies on the beach, with their heads facing the center point.



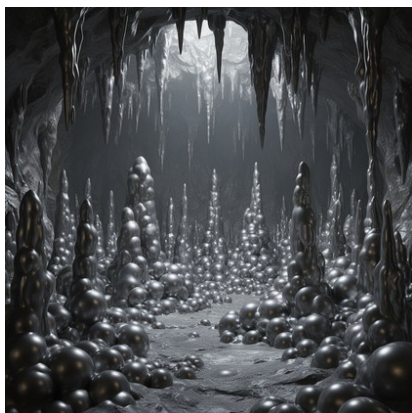
single pass
× 1/3 Attribute - Shape, Grammar - Consistency



Kimi K3
× 2/3 Grammar - Consistency
FLUX.2-klein-9B



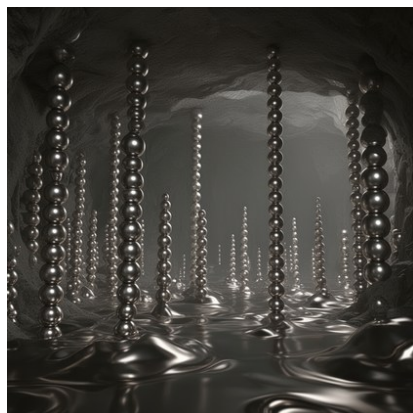
Qwen3.8-27B
× 2/3 Grammar - Consistency



single pass
× 0/3 Attribute - Shape, Grammar - Consistency,
Compound - Imagination



Kimi K3
✓ 3/3
FLUX.2-dev



Qwen3.8-27B
× 1/3 Grammar - Consistency, Compound -
Imagination

052

Figure 3. A cave scene sculpture made entirely of flowing liquid metal, in which all stalactites and stalagmites take on a uniform, smooth spherical shape.



single pass
× 0/2 Compound - Feature Matching, Logical Reasoning



Kimi K3
✓ 2/2

FLUX.2-klein-9B



Qwen3.8-27B
✓ 2/2

053



single pass
× 0/2 Compound - Feature Matching, Logical Reasoning



Kimi K3
✓ 2/2

FLUX.2-dev



Qwen3.8-27B
✓ 2/2

Figure 4. In an empty park, the ground and all rides are dry, except for a pool of water on the seat of a swing.



single pass
✗ 0/1 Entity Layout - Three-Dimensional Space



Kimi K3
✓ 1/1
FLUX.2-klein-9B



Qwen3.8-27B
✓ 1/1



single pass
✗ 0/1 Entity Layout - Three-Dimensional Space



Kimi K3
✓ 1/1
FLUX.2-dev



Qwen3.8-27B
✓ 1/1

054

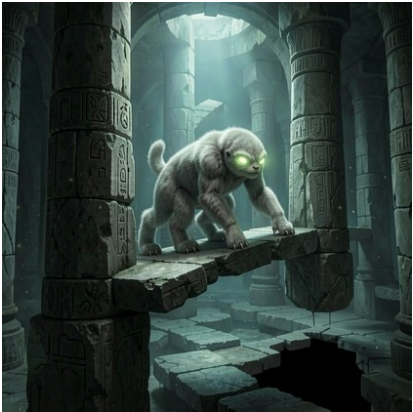
Figure 5. A tower of smooth cobblestones suspended above the calm water, and beyond is a vague mountain, the entire scene assumes a Zen feel.



single pass
✓ 2/2

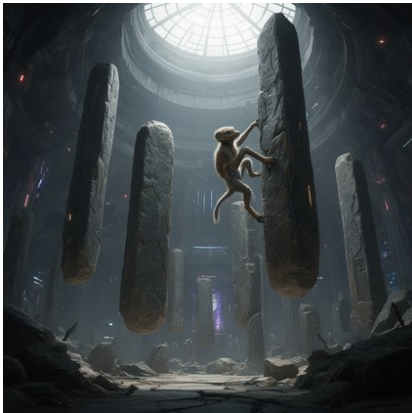


Kimi K3
✓ 2/2
FLUX.2-klein-9B



Qwen3.8-27B
✓ 2/2

055



single pass
✗ 0/2 Entity Layout - Three-Dimensional Space,
Action - Animal



Kimi K3
✓ 2/2
FLUX.2-dev



Qwen3.8-27B
✓ 2/2

Figure 6. Inside the huge alien relic, a furry six-legged alien creature is carefully climbing from one huge floating stone pillar to another.



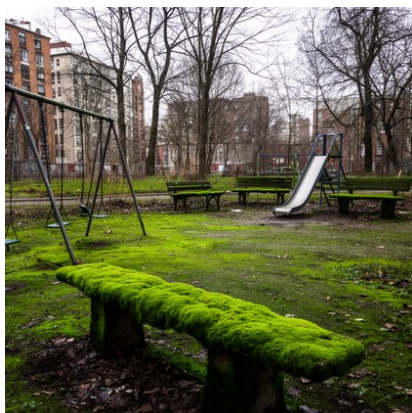
single pass
✗ 0/1 Grammar - Consistency



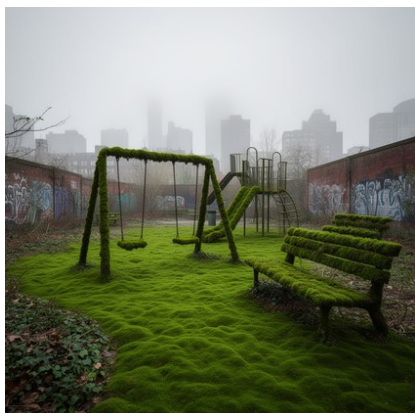
Kimi K3
✓ 1/1
FLUX.2-klein-9B



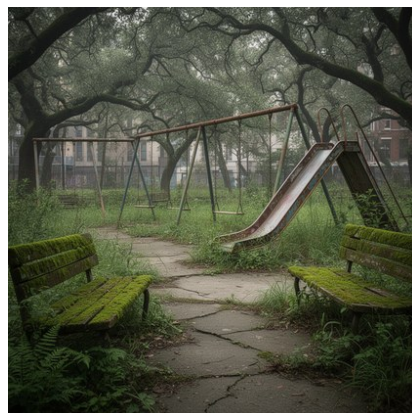
Qwen3.8-27B
✓ 1/1



single pass
✗ 0/1 Grammar - Consistency



Kimi K3
✓ 1/1
FLUX.2-dev



Qwen3.8-27B
✗ 0/1 Grammar - Consistency

056

Figure 7. An abandoned city park with swings, slides and benches covered in thick, emerald green moss.



single pass
✗ 0/2 Grammar - Pronoun Reference, Style



Kimi K3
✓ 2/2
FLUX.2-klein-9B



Qwen3.8-27B
✓ 2/2

057



single pass
✓ 2/2



Kimi K3
✓ 2/2
FLUX.2-dev



Qwen3.8-27B
✓ 2/2

Figure 8. A potato astronaut in a spacesuit is watering a rose plant on the moon. It looks happy, Cyberpunk style.



single pass
✗ 0/2 Logical Reasoning, Action - Full-body
(Character/Anthropomorphic)



Kimi K3
✓ 2/2
FLUX.2-klein-9B



Qwen3.8-27B
✓ 2/2



single pass
✗ 0/2 Logical Reasoning, Action - Full-body
(Character/Anthropomorphic)



Kimi K3
✓ 2/2
FLUX.2-dev



Qwen3.8-27B
✓ 2/2

058

Figure 9. A little yellow duck wearing a raincoat stood in front of a puddle. Instead of jumping in and playing in the water, it opened a small umbrella.



single pass
✓ 2/2



Kimi K3
✓ 2/2
FLUX.2-klein-9B



Qwen3.8-27B
✓ 2/2

059



single pass
✗ 0/2 Relationship - Composition, Attribute - Quantity



Kimi K3
✓ 2/2
FLUX.2-dev



Qwen3.8-27B
✓ 2/2

Figure 10. On the park bench at dusk, there are three books and two apples. One of the books is open and the apple is on the left side of the book.



single pass
✗ 0/1 Style



Kimi K3
✓ 1/1
FLUX.2-klein-9B



Qwen3.8-27B
✓ 1/1



single pass
✗ 0/1 Style



Kimi K3
✓ 1/1
FLUX.2-dev



Qwen3.8-27B
✗ 0/1 Style

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Figure 11. In the Impressionist oil painting style, a fashion designer's workbench is littered with fabrics and sketches, and a freshly made evening dress hangs on a mannequin next to it.

061 C. Tool Schemata

062 Tools are exposed to the agent using JSON schemata, following the MCP [4] tool convention [6]. This appendix gives
063 compact parameter summaries derived from the implementation, using descriptive media-type names. Complete machine-
064 readable tool definitions, including shared and nested definitions, are exported by the MCP SDK in `code/tool-schemas.json`.

065 `generate_image`

```
066 {  
067   "name": "generate_image",  
068   "description": "Generate a new image from a text prompt.",  
069   "inputSchema": {  
070     "type": "object",  
071     "properties": {  
072       "prompt": {  
073         "type": "string",  
074         "description": "text description of the image to generate."  
075       },  
076       "width": {  
077         "type": "integer",  
078         "description": "output width in pixels (default 1024).",  
079         "default": 1024  
080       },  
081       "height": {  
082         "type": "integer",  
083         "description": "output height in pixels (default 1024).",  
084         "default": 1024  
085       },  
086       "steps": {  
087         "type": "integer",  
088         "description": "diffusion steps; omit for the model default."  
089       },  
090       "guidance": {  
091         "type": "number",  
092         "description": "guidance scale; omit for the model default."  
093       },  
094       "seed": {  
095         "type": "integer",  
096         "description": "random seed; randomized if omitted."  
097       },  
098       "model": {  
099         "type": "string",  
100         "enum": [  
101           "flux2-dev",  
102           "flux2-klein",  
103           "flux1-dev",  
104           "qwen-image"  
105         ],  
106         "description": "generation model (default flux2-dev \u2014 medium cost, more  
107           creative). flux2-klein and qwen-image are cheap with high prompt  
108           adherence. flux1-dev is a weaker/older base model but has by far the  
109           largest LoRA ecosystem \u2014 reach for it when you need a specific  
110           adapter, not for base quality. flux2-klein/qwen-image/flux1-dev take  
111           LoRAs; flux2-dev does not.",
```

```
    "default": "DEFAULT_GENERATE_MODEL"
  },
  "loras": {
    "type": "object[]",
    "description": "LoRA adapters to stack. flux2-klein takes FLUX.2-klein-9B
      adapters and qwen-image takes Qwen-Image adapters; flux2-dev does NOT
      support LoRAs (raises ValueError). Put any trigger words in the prompt."
  }
},
"required": [
  "prompt"
]
}
```

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```
126 edit_image
127 {
128   "name": "edit_image",
129   "description": "Edit an image using a text instruction (whole-image, no mask \
130     u2014 for masked/region edits use 'inpaint_image').",
131   "inputSchema": {
132     "type": "object",
133     "properties": {
134       "image": {
135         "type": "image",
136         "description": "image to edit (ImageContent).",
137       },
138       "prompt": {
139         "type": "string",
140         "description": "the edit instruction."
141       },
142       "references": {
143         "type": "image[]",
144         "description": "additional images (ImageContent) whose visual details the
145           edit should preserve or borrow."
146       },
147       "steps": {
148         "type": "integer",
149         "description": "diffusion steps; omit for the model default."
150       },
151       "guidance": {
152         "type": "number",
153         "description": "guidance scale; omit for the model default."
154       },
155       "seed": {
156         "type": "integer",
157         "description": "random seed; randomized if omitted."
158       },
159       "width": {
160         "type": "integer",
161         "description": "output width; omit for the model default."
162       },
163       "height": {
164         "type": "integer",
165         "description": "output height; omit for the model default."
166       },
167       "model": {
168         "type": "string",
169         "enum": [
170           "firered",
171           "flux2-dev",
172           "flux2-klein",
173           "qwen-image"
174         ],
175         "description": "editing model (default firered \u2014 cheap, good for most
176           edits). qwen-image and flux2-klein are also cheap/high adherence; flux2-
177           dev is medium cost, more creative. firered/qwen-image/flux2-klein take
178           LoRAs; flux2-dev does not.",
```

```
    "default": "DEFAULT_EDIT_MODEL" 179
  }, 180
  "loras": { 181
    "type": "object[]", 182
    "description": "LoRA adapters to stack. qwen-image takes Qwen-Image-Edit 183
      -2511 adapters (the true base \u2014 prefer it when LoRA fidelity 184
      matters); firered loads the same adapters but, as a finetune, may weaken 185
      their effect; flux2-klein takes FLUX.2-klein-9B adapters; flux2-dev 186
      does NOT support LoRAs (raises ValueError). Put any trigger words in the 187
      prompt." 188
  } 189
}, 190
"required": [ 191
  "image", 192
  "prompt" 193
] 194
} 195
} 196
```



```
197 inpaint_image
198 {
199   "name": "inpaint_image",
200   "description": "Inpaint the masked region of an image with FLUX.1 Fill.",
201   "inputSchema": {
202     "type": "object",
203     "properties": {
204       "image": {
205         "type": "image",
206         "description": "image to inpaint (ImageContent).",
207       },
208       "mask": {
209         "type": "image",
210         "description": "grayscale mask (ImageContent); white = regenerate, black =
211           preserve.",
212       },
213       "prompt": {
214         "type": "string",
215         "description": "description of the desired contents of the masked region."
216       },
217       "steps": {
218         "type": "integer",
219         "description": "diffusion steps (default 50).",
220       },
221       "guidance": {
222         "type": "number",
223         "description": "guidance scale (default 30.0).",
224       },
225       "seed": {
226         "type": "integer",
227         "description": "random seed; randomized if omitted.",
228       },
229       "dilate": {
230         "type": "integer",
231         "description": "grow the fill region by ~N px before inpainting \u2014 use
232           10-30 for shape-changing/removal edits so the mask silhouette doesn't
233           anchor the old shape, and describe what replaces the removed parts (e.g.
234           'bare arms').",
235         "default": 0
236       },
237       "loras": {
238         "type": "object[]",
239         "description": "FLUX.1-dev LoRA adapters (FLUX.1 Fill takes FLUX.1-dev
240           adapters). Put any trigger words in the prompt."
241       }
242     },
243     "required": [
244       "image",
245       "mask",
246       "prompt"
247     ]
248   }
249 }
```

outpaint_image	250
{	251
"name": "outpaint_image",	252
"description": "Extend an image beyond its borders (outpainting) with FLUX.1	253
Fill.",	254
"inputSchema": {	255
"type": "object",	256
"properties": {	257
"image": {	258
"type": "image",	259
"description": "image to extend (ImageContent)."	260
},	261
"prompt": {	262
"type": "string",	263
"description": "description of the full scene being revealed. left/right/top	264
/bottom: pixels to add on each side; at least one side must be > 0."	265
},	266
"left": {	267
"type": "integer",	268
"default": 0	269
},	270
"right": {	271
"type": "integer",	272
"default": 0	273
},	274
"top": {	275
"type": "integer",	276
"default": 0	277
},	278
"bottom": {	279
"type": "integer",	280
"default": 0	281
},	282
"overlap": {	283
"type": "integer",	284
"description": "pixels of the original border regenerated on padded sides so	285
tone and structures blend across the transition (0 disables; risks a	286
visible seam along the image edge).",	287
"default": 20	288
},	289
"steps": {	290
"type": "integer",	291
"description": "diffusion steps (default 50)."	292
},	293
"guidance": {	294
"type": "number",	295
"description": "guidance scale (default 30.0)."	296
},	297
"seed": {	298
"type": "integer",	299
"description": "random seed; randomized if omitted."	300
},	301
"loras": {	302

```
303         "type": "object[]",
304         "description": "FLUX.1-dev LoRA adapters. Put any trigger words in the
305             prompt."
306     }
307 },
308 "required": [
309     "image",
310     "prompt"
311 ]
312 }
313 }
```

```
inspect_region 314
{ 315
  "name": "inspect_region", 316
  "description": "Crop a region out of an image for full-resolution visual 317
    inspection.", 318
  "inputSchema": { 319
    "type": "object", 320
    "properties": { 321
      "image": { 322
        "type": "image", 323
        "description": "image to crop from (ImageContent).", 324
      }, 325
      "roi": { 326
        "type": "InspectRegionRoi", 327
        "description": "region to crop. Pass a text description such as 'the logo' 328
          to segment with SAM, a rectangle as [x1, y1, x2, y2], or an object with 329
          x/y/width/height or x1/y1/x2/y2. Rectangle coordinates are pixels unless 330
          normalized=true.", 331
      }, 332
      "normalized": { 333
        "type": "boolean", 334
        "description": "treat rectangle coordinates as 0-1 fractions of image width/ 335
          height instead of absolute pixels.", 336
        "default": false 337
      }, 338
      "padding": { 339
        "type": "number", 340
        "description": "extra context around the crop \u2014 pixels, or a fraction 341
          of each crop dimension when between 0 and 1.", 342
        "default": 0.0 343
      }, 344
      "min_size": { 345
        "type": "integer", 346
        "description": "minimum crop side length in pixels.", 347
        "default": 1 348
      }, 349
      "threshold": { 350
        "type": "number", 351
        "description": "SAM3 instance filter (text roi only).", 352
        "default": 0.3 353
      }, 354
      "mask_threshold": { 355
        "type": "number", 356
        "description": "SAM3 mask binarization (text roi only).", 357
        "default": 0.5 358
      } 359
    }, 360
    "required": [ 361
      "image", 362
      "roi" 363
    ] 364
  } 365
}
```

```
367 segment
368 {
369   "name": "segment",
370   "description": "Run text-driven concept segmentation (SAM3 PCS) on an image.",
371   "inputSchema": {
372     "type": "object",
373     "properties": {
374       "image": {
375         "type": "image",
376         "description": "image to segment (ImageContent).",
377       },
378       "description": {
379         "type": "string",
380         "description": "text description of the region to segment, e.g. \"all
381           clothing\" or \"the sky\".",
382       },
383       "threshold": {
384         "type": "number",
385         "description": "filters detected instances.",
386         "default": 0.3
387       },
388       "mask_threshold": {
389         "type": "number",
390         "description": "binarizes masks.",
391         "default": 0.5
392       },
393       "negative_descriptions": {
394         "type": "list[str]",
395         "description": "segmented separately and subtracted from the positive mask."
396       }
397     },
398     "required": [
399       "image",
400       "description"
401     ]
402   }
403 }
```



```
segment_points 404
{ 405
  "name": "segment_points", 406
  "description": "Run point-prompt segmentation (SAM3 Tracker / PVS) on an image 407
  .", 408
  "inputSchema": { 409
    "type": "object", 410
    "properties": { 411
      "image": { 412
        "type": "image", 413
        "description": "image to segment (ImageContent).", 414
      }, 415
      "points": { 416
        "type": "list[list[float]]", 417
        "description": "positive [x, y] clicks for one target.", 418
      }, 419
      "negative_points": { 420
        "type": "list[list[float]]", 421
        "description": "negative clicks for refinement.", 422
      }, 423
      "labels": { 424
        "type": "list[int]", 425
        "description": "overrides point labels directly (1=positive, 0=negative).", 426
      }, 427
      "normalized": { 428
        "type": "boolean", 429
        "description": "treat coordinates as 0-1 fractions of image width/height 430
        instead of absolute pixels.", 431
        "default": false 432
      }, 433
      "multimask_output": { 434
        "type": "boolean", 435
        "description": "request candidate masks; by default the highest-IoU 436
        candidate is selected.", 437
        "default": true 438
      }, 439
      "mask_index": { 440
        "type": "integer", 441
        "description": "force a specific candidate mask.", 442
      }, 443
      "mask_threshold": { 444
        "type": "number", 445
        "description": "mask decoding threshold.", 446
        "default": 0.0 447
      }, 448
      "max_hole_area": { 449
        "type": "number", 450
        "description": "fill holes up to this area in post-processing.", 451
        "default": 0.0 452
      }, 453
      "max_sprinkle_area": { 454
        "type": "number", 455
        "description": "remove specks up to this area.", 456
```

```
457         "default": 0.0
458     },
459     "apply_non_overlapping_constraints": {
460         "type": "boolean",
461         "description": "forwarded to SAM3 tracker post-processing.",
462         "default": false
463     }
464 },
465 "required": [
466     "image",
467     "points"
468 ]
469 }
470 }
```

```
segment_boxes 471
{ 472
  "name": "segment_boxes", 473
  "description": "Run box-prompt segmentation (SAM3 Tracker / PVS) on an image.", 474
  "inputSchema": { 475
    "type": "object", 476
    "properties": { 477
      "image": { 478
        "type": "image", 479
        "description": "image to segment (ImageContent).", 480
      }, 481
      "boxes": { 482
        "type": "list[list[float]]", 483
        "description": "[x1, y1, x2, y2] boxes; all resulting object masks are 484
          merged." 485
      }, 486
      "normalized": { 487
        "type": "boolean", 488
        "description": "treat coordinates as 0-1 fractions of image width/height 489
          instead of absolute pixels.", 490
        "default": false 491
      }, 492
      "multimask_output": { 493
        "type": "boolean", 494
        "description": "request candidate masks per box.", 495
        "default": false 496
      }, 497
      "mask_index": { 498
        "type": "integer", 499
        "description": "force a specific candidate mask." 500
      }, 501
      "mask_threshold": { 502
        "type": "number", 503
        "description": "mask decoding threshold.", 504
        "default": 0.0 505
      }, 506
      "max_hole_area": { 507
        "type": "number", 508
        "description": "fill holes up to this area in post-processing.", 509
        "default": 0.0 510
      }, 511
      "max_sprinkle_area": { 512
        "type": "number", 513
        "description": "remove specks up to this area.", 514
        "default": 0.0 515
      }, 516
      "apply_non_overlapping_constraints": { 517
        "type": "boolean", 518
        "description": "forwarded to SAM3 tracker post-processing.", 519
        "default": false 520
      } 521
    }, 522
    "required": [ 523
```

```
524         "image",  
525         "boxes"  
526     ]  
527 }  
528 }
```

```
invert_mask 529
{ 530
  "name": "invert_mask", 531
  "description": "Invert a mask (255 \u2194 0). Pure pixel operation \u2014 no 532
    model loaded.", 533
  "inputSchema": { 534
    "type": "object", 535
    "properties": { 536
      "mask": { 537
        "type": "image", 538
        "description": "grayscale mask to invert (ImageContent).\" 539
      } 540
    }, 541
    "required": [ 542
      "mask" 543
    ] 544
  } 545
}
```



```
547 upscale_image
548 {
549   "name": "upscale_image",
550   "description": "Super-resolution upscale of an image (FaithDiff).",
551   "inputSchema": {
552     "type": "object",
553     "properties": {
554       "image": {
555         "type": "image",
556         "description": "image to upscale (ImageContent).",
557       },
558       "scale": {
559         "type": "number",
560         "description": "size multiplier like 2 or 4.",
561       },
562       "target_width": {
563         "type": "integer",
564         "description": "explicit output width in pixels.",
565       },
566       "target_height": {
567         "type": "integer",
568         "description": "explicit output height in pixels.",
569       },
570       "steps": {
571         "type": "integer",
572         "description": "diffusion steps per tile.",
573         "default": 20
574       },
575       "guidance": {
576         "type": "number",
577         "description": "guidance scale.",
578         "default": 5.0
579       },
580       "target_tile_size": {
581         "type": "integer",
582         "description": "SR tile size in pixels.",
583         "default": 1024
584       },
585       "overlap": {
586         "type": "number",
587         "description": "tile overlap fraction.",
588         "default": 0.5
589       }
590     },
591     "required": [
592       "image"
593     ]
594   }
595 }
```

```
generate_video 596
{ 597
  "name": "generate_video", 598
  "description": "Generate a video from an image using Wan 2.2.", 599
  "inputSchema": { 600
    "type": "object", 601
    "properties": { 602
      "image": { 603
        "type": "image", 604
        "description": "first frame (ImageContent).", 605
      }, 606
      "prompt": { 607
        "type": "string", 608
        "description": "text description of the desired motion/scene.", 609
      }, 610
      "last_image": { 611
        "type": "image", 612
        "description": "optional last frame (ImageContent) for FLF2V.", 613
      }, 614
      "duration_seconds": { 615
        "type": "number", 616
        "description": "clip length at the model-native 16 fps.", 617
        "default": 5.0 618
      }, 619
      "steps": { 620
        "type": "integer", 621
        "description": "diffusion steps.", 622
        "default": 30 623
      }, 624
      "guidance": { 625
        "type": "number", 626
        "description": "guidance scale.", 627
        "default": 3.5 628
      }, 629
      "guidance_2": { 630
        "type": "number", 631
        "description": "optional second-stage guidance (Wan MoE).", 632
      }, 633
      "flow_shift": { 634
        "type": "number", 635
        "description": "Wan scheduler flow shift.", 636
        "default": 5.0 637
      }, 638
      "seed": { 639
        "type": "integer", 640
        "description": "random seed; randomized if omitted.", 641
      }, 642
      "negative_prompt": { 643
        "type": "string", 644
        "description": "overrides the default Wan negative prompt.", 645
      }, 646
      "short_edge": { 647
        "type": "integer", 648
```

```
649         "description": "output short-edge resolution.",
650         "default": 720
651     },
652     "num_frames": {
653         "type": "integer",
654         "description": "overrides duration_seconds with an exact count."
655     }
656 },
657 "required": [
658     "image",
659     "prompt"
660 ]
661 }
662 }
```

```
interpolate_video 663
{ 664
  "name": "interpolate_video", 665
  "description": "Frame-interpolate a video by inserting synthesized frames 666
    between each consecutive pair (EMA-VFI).", 667
  "inputSchema": { 668
    "type": "object", 669
    "properties": { 670
      "video": { 671
        "type": "EmbeddedResource", 672
        "description": "the EmbeddedResource a video tool returned. Pass it through 673
          unchanged, as you would an ImageContent between image tools." 674
      }, 675
      "n_insert": { 676
        "type": "integer", 677
        "description": "frames to insert between each pair (must be >= 1; use 1 for 678
          the cheap doubling case).", 679
        "default": 2 680
      }, 681
      "use_tta": { 682
        "type": "boolean", 683
        "description": "toggles EMA-VFI test-time augmentation.", 684
        "default": false 685
      } 686
    }, 687
    "required": [ 688
      "video" 689
    ] 690
  } 691
}
```

```
693 upscale_video
694 {
695   "name": "upscale_video",
696   "description": "Super-resolution every frame of a clip (FaithDiff), at the
697     source fps.",
698   "inputSchema": {
699     "type": "object",
700     "properties": {
701       "video": {
702         "type": "EmbeddedResource",
703         "description": "the clip to upscale (the EmbeddedResource a video tool
704           returned). Pass it through unchanged."
705       },
706       "scale": {
707         "type": "number",
708         "description": "size multiplier like 2 or 4."
709       },
710       "target_width": {
711         "type": "integer",
712         "description": "explicit output width in pixels."
713       },
714       "target_height": {
715         "type": "integer",
716         "description": "explicit output height in pixels."
717       },
718       "steps": {
719         "type": "integer",
720         "description": "diffusion steps per tile.",
721         "default": 20
722       },
723       "guidance": {
724         "type": "number",
725         "description": "guidance scale.",
726         "default": 5.0
727       },
728       "target_tile_size": {
729         "type": "integer",
730         "description": "SR tile size in pixels.",
731         "default": 1024
732       },
733       "overlap": {
734         "type": "number",
735         "description": "tile overlap fraction.",
736         "default": 0.5
737       }
738     },
739     "required": [
740       "video"
741     ]
742   }
743 }
```

D. Instructions and System Prompts

For reproducibility and transparency we also show the exact system prompts we use. This includes the MCP [4] instructions field which is sent to the agent (via the harness) from the MCP server and the AGENTS.md that we used for the UniGen-Bench++ [7] experiment that is read by the batch harness.

The full Klein policy is reproduced below. The code bundle also contains AGENTS.flux2-dev.md and AGENTS.qwen-image.md, which provide the generator-specific model instructions used for the other two arms while retaining the same inspection, editing, and stopping procedure.

MCP Instructions

Continuum image/video model server.

Inputs / outputs: image inputs are MCP ImageContent objects ({type: 'image', data: base64 PNG/JPEG/etc., mimeType: 'image/png'}); video inputs are encoded MP4 bytes. Outputs are inline image content or AV1 MP4 video bytes.

Editing discipline

These rules govern HOW to call the editing tools and materially affect output quality. They came from manual and agent-driven testing against FLUX.2 + SAM3.

edit_image prompt rules

- **Delta, not cumulative.** Each call starts from the current image. Describe only the CHANGE you want, not what already exists. Write "make the dog larger" -- not "a golden retriever lying on grass, dog is larger".
- **Concrete visual language, not vague adjectives.** FLUX.2 responds to noun-phrase descriptions. "warm golden-hour rim lighting on the left side of the face" beats "make it more dramatic". Avoid words like "better", "nicer", "more interesting".
- **Removals: describe the fill, not the removal.** "Fill with the surrounding tile pattern" beats "remove the trash can". The model needs to know what should be there, not what shouldn't.
- **Use references to lock visual aspects.** When an edit risks changing something that was already correct, pass a prior version's URI in 'references' and write the prompt as preserving that look.

edit_image vs inpaint_image: default to edit_image

- **edit_image with a targeted instruction is the default** -- even for localized changes. Despite the apparent lack of control, a precise "replace X with Y" instruction ("replace the red sedan with a blue bicycle") usually localizes well, preserves the rest of the image, and produces better quality than masked inpainting. Name what to replace AND what it should become.
- **inpaint_image** regenerates only the white region of a 'mask' and preserves the rest pixel-exact. Reserve it for specific situations: pixel-exact preservation outside the region is a hard requirement; edit_image keeps leaking changes outside the target after 2-3 attempts; or the region is only expressible as a precise mask (e.g. invert_mask "everything except X" edits).


```
795 - Over-masking is more harmful than under-masking; when in doubt,  
796 edit_image first.  
797  
798 ## When you do inpaint (segmentation + inpaint_image)  
799  
800 - Segment first, then inpaint_image with the mask URI. Use text  
801 'segment' for semantic classes ("all clothing", "the sky"),  
802 'segment_boxes' when you can identify a bounding box, and  
803 'segment_points' for precise clicks/refinement.  
804 - **Negative grounding:** for text masks, pass 'negative_descriptions'.  
805 For visual masks, pass 'negative_points' or explicit point 'labels'.  
806 - **Prompt describes the CONTENTS of the region**, not an edit  
807 instruction: "a red silk shirt", not "make it red". Do NOT mention  
808 the mask or the region itself in the prompt.  
809 - **Removing or reshaping an object** (long sleeves → sleeveless,  
810 hat removed, smaller bag): a mask that hugs the old silhouette will  
811 regenerate the same shape. Set 'dilate' (10-30 px) to give the fill  
812 room, and describe what replaces the removed parts ("bare arms and  
813 shoulders"), not just the new object.  
814 - For "remove X" or "edit everything except X": use segment to mask X,  
815 then invert_mask, then inpaint_image with the inverted mask.  
816 - **Know inpaint's limits.** The fill is text-driven only: it cannot  
817 reproduce a specific object or person from another image (use 'loras'  
818 for trained subjects, or edit_image with references when the whole  
819 image may legitimately change). Structural changes much larger than  
820 the mask are unreliable, and small regions render with limited fine  
821 detail. If 2-3 seeds fail, change the mask or the approach instead of  
822 iterating on the prompt.  
823 - **Extending the canvas** (wider shot, more sky, re-aspect): use  
824 outpaint_image with per-side pixel padding -- do NOT build the padded  
825 canvas and mask yourself. The prompt describes the full scene being  
826 revealed.  
827  
828 ## LoRA support  
829  
830 Which tools/models accept 'loras', and which adapter family each needs:  
831  
832 - generate_image: 'flux1-dev' (FLUX.1-dev adapters -- by far the largest  
833 and most reliable LoRA ecosystem, but a weaker/older base model, so  
834 reach for it for its adapters, not for base quality), 'flux2-klein'  
835 (FLUX.2-klein-9B adapters), and 'qwen-image' (Qwen-Image adapters).  
836 'flux2-dev' does NOT support LoRAs.  
837 - edit_image: 'qwen-image' (Qwen-Image-Edit-2511 adapters, the actual  
838 base checkpoint) and 'flux2-klein' (FLUX.2-klein-9B adapters).  
839 'firered' accepts Qwen-Image-Edit adapters too, but it is a finetune --  
840 adapters load yet their effect may be weakened; prefer  
841 model='qwen-image' when LoRA fidelity matters. 'flux2-dev' does NOT  
842 support LoRAs.  
843 - inpaint_image / outpaint_image: always FLUX.1 Fill -- FLUX.1-dev  
844 family adapters.  
845 - Video and upscale tools take no LoRAs.  
846  
847 Failure modes: passing 'loras' with an unsupported model raises an
```

error naming the LoRA-capable models. An adapter trained for a DIFFERENT base model fails at load time -- or in the worst case loads as a silent no-op; if a LoRA appears to do nothing, suspect a base-model mismatch before adjusting 'scale'. Find adapters on Hugging Face by base-model tag, e.g. 'base_model:adapter:black-forest-labs/FLUX.1-dev'. Trigger words must be written into the prompt yourself. On first use a new adapter emits "downloading LoRA" then "applying LoRA" progress; a cached adapter skips straight to applying.

Steps and guidance

Omit 'steps' and 'guidance' unless you have a specific reason -- each model ships defaults tuned to its own regime, and overriding them blind usually costs quality. The defaults:

- **flux2-dev** (generate and edit): ~30 steps, guidance ~4.
- **flux1-dev** (generate only): ~28 steps, guidance ~3.5.
- **qwen-image**: 50 steps for generate / 40 for edit, guidance ~4.
- **firered** edit: ~30 steps, guidance ~4.
- **flux2-klein** (generate and edit): **4 steps, guidance 1.0**. klein is a distilled model, so the tiny step count and guidance near 1 are intrinsic, not a cheap default. Raising guidance fights the distillation (expect oversaturation and artifacts) and extra steps buy almost nothing -- leave both alone.
- **inpaint_image / outpaint_image** (FLUX.1 Fill): 50 steps, **guidance ~30**. That high guidance is correct for Fill, not a typo -- do not drag it down toward the ~4 the other models use, which under-conditions the fill.

When to inspect a crop

- **Fine details** (logos, small text, eye direction, artifacts, hands, jewelry, edges): call 'inspect_region' with a rectangle or text description before deciding. Full images may be downsampled by VLMs; 'inspect_region' returns the crop at source resolution.

Self-critique loop

After each edit, evaluate the result against the user's intent before deciding what to do next:

- **Approach right, specifics wrong** (artifact, off composition, wrong expression, unwanted color shift): call edit_image again with the same prompt and a different 'seed'. Try 2-3 seeds before giving up on the prompt.
- **Approach wrong**: do not iterate with edit_image. Rethink. Maybe the prompt needs to be reformulated, or the change should be a localized inpaint_image, or it's actually two edits chained.
- **Result is good**: stop calling tools and tell the user. Don't over-edit good results.
- **Several rounds in, quality drifting**: pass the earliest good-looking version's URI in 'references' and explicitly mention preserving its qualities in the prompt.

```
901  ## Multi-step composition
902
903  For complex prompts that mention multiple distinct changes (e.g.
904  "make the dog smaller, change the sky to sunset, add a kite"), split
905  into ordered single-entity edits rather than one composite prompt.
906  FLUX handles one change per call cleanly; multiple changes per prompt
907  produce attribute leakage between entities.
```

AGENTS.md	908
# Image Generator -- benchmark policy	909
You generate an image for a given prompt, then improve it until it matches.	910
## Model	911
Always use <code>**flux2-klein**</code> for generation and <code>**flux2-klein**</code> for editing.	912
This is fixed. The baseline arm of this experiment calls the same model directly, so changing it here would measure a model swap instead of the value of iterating.	913
flux2-klein also serves both generation and editing from one loaded model, so alternating between them costs nothing -- which matters when a single run may alternate many times.	914
Pass <code>'model: "flux2-klein"'</code> on <code>**every**</code> <code>'generate_image'</code> and <code>'edit_image'</code> call. Omitting it does not use the model named above -- the server falls back to its own defaults, which are a different generator and a different editor, and the run completes without saying so.	915
Omit <code>'steps'</code> and <code>'guidance'</code> ; the model's own defaults (4 steps, guidance 1.0) are correct for it.	916
## Procedure	917
<code>'generate'</code> → <code>'inspect'</code> → <code>'edit'</code> , repeating the last two while the image is clearly wrong. Mark your best result as you go, so an interrupted run still returns something you chose.	918
### Generate	919
Generate from the user's description. Fill in <code>*visual*</code> detail the prompt implies, but add nothing non-visual -- no names, no backstory. The first generation is usually imperfect.	920
### Inspect	921
Look at what you actually produced, not what you intended. Reason about it concretely before deciding anything.	922
<code>**When the prompt asks for rendered text, verify the text by zooming.**</code> Use <code>'inspect_region'</code> on each area that should contain words: at full-image scale the glyphs are downsampled and wrong spellings look plausible. Read the letters one by one and compare with the requested string, including capitalisation.	923
### Edit	924
Make targeted edits naming only the part that is wrong. Prefer <code>'edit_image'</code> ; reach for <code>'inpaint_image'</code> when you need a change confined to one region.	925
Do not try to widen the frame. The image size is fixed for comparison, so	926

961 enlarging the canvas makes the result unscorable -- if the subject does not fit,
962 regenerate with a composition that does.

963

964 For text specifically: name the incorrect string and the correct one. Do not
965 re-describe the rest of the scene, or it will drift.

966

967 ### Mark your best

968

969 As soon as you have an image you would be willing to submit, call 'mark_best'
970 on it. Call it again whenever a later image is better. Only one image is marked
971 at a time, and the most recent call wins.

972

973 **Your last action before stopping is to check the mark.** If you edited
974 anything after marking, either mark the newer image or say why you are keeping
975 the older one. A mark left behind by later work returns the version from before
976 your final fix -- observed in an earlier run, where the closing summary described
977 changes the marked image did not contain.

978

979 Mark early and often, because **the marked image is what gets returned if the
980 run is cut short** -- by the step limit, or by an error you cannot see coming.
981 An unmarked run that is interrupted returns whatever happened to be produced
982 last, which may be a version you had already rejected.

983

984 Marking does not stop the run. Mark, then carry on improving.

985

986 Afterwards you can pass 'best' anywhere an image id is accepted, so
987 'edit_image(image="best")' continues from your marked image without tracking
988 its id. That matters most when the image you marked is *not* the newest one.

989

990 ## Stopping

991

992 Stop when every element the prompt asks for is present and correct, and the
993 image you marked is the one you want returned. **Good enough is a stop
994 condition.** Do not keep editing to chase a marginal
995 improvement, and do not restart from scratch to escape a small flaw.

996

997 Two failure modes to avoid, both observed in earlier runs:

998

999 - **Destroying a good result.** If an edit makes things worse, 'mark_best' the
1000 earlier image and continue from it with 'image="best"'. Never regenerate from
1001 scratch when you already have an image that mostly works -- you will lose it.

1002 - **Chasing the impossible.** If several different attempts fail to fix the
1003 same thing, the model probably cannot render it. Mark the best version you
1004 have and stop. Twenty steps spent on one detail is a failure, not diligence.

1005

1006 ## Notes

1007

1008 - Do not set a seed. Sizes and seeds are controlled by the harness.

1009 - You may use previously generated images as references where the model
1010 supports it. If you do, tell the editor to take a specific feature from the
1011 reference -- do not describe the reference's contents.

References1012

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